Sprawozdanie LAB 4

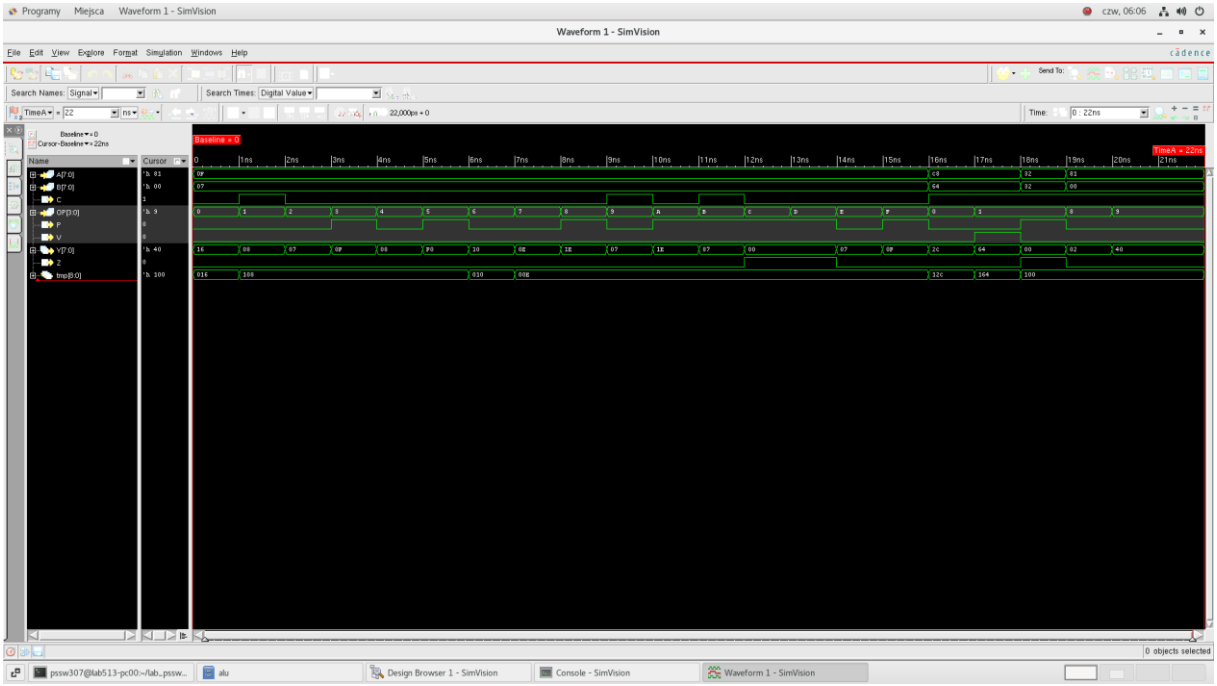
Jakub Lasak

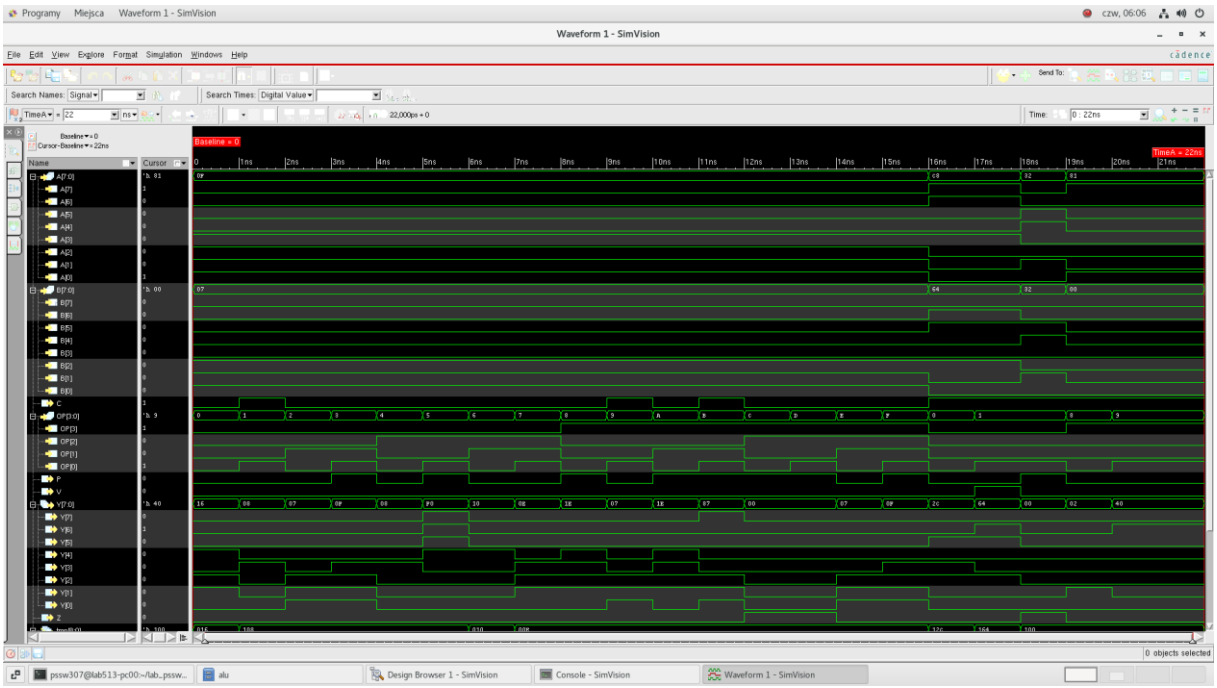
Pssw307

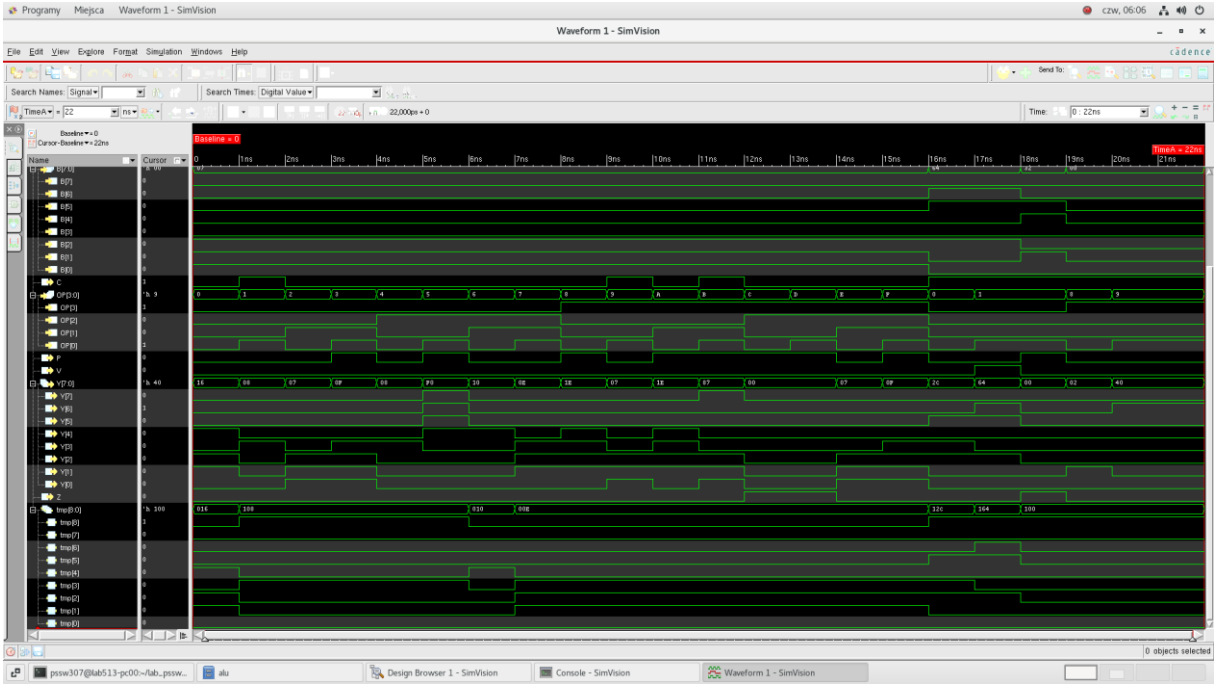
412065

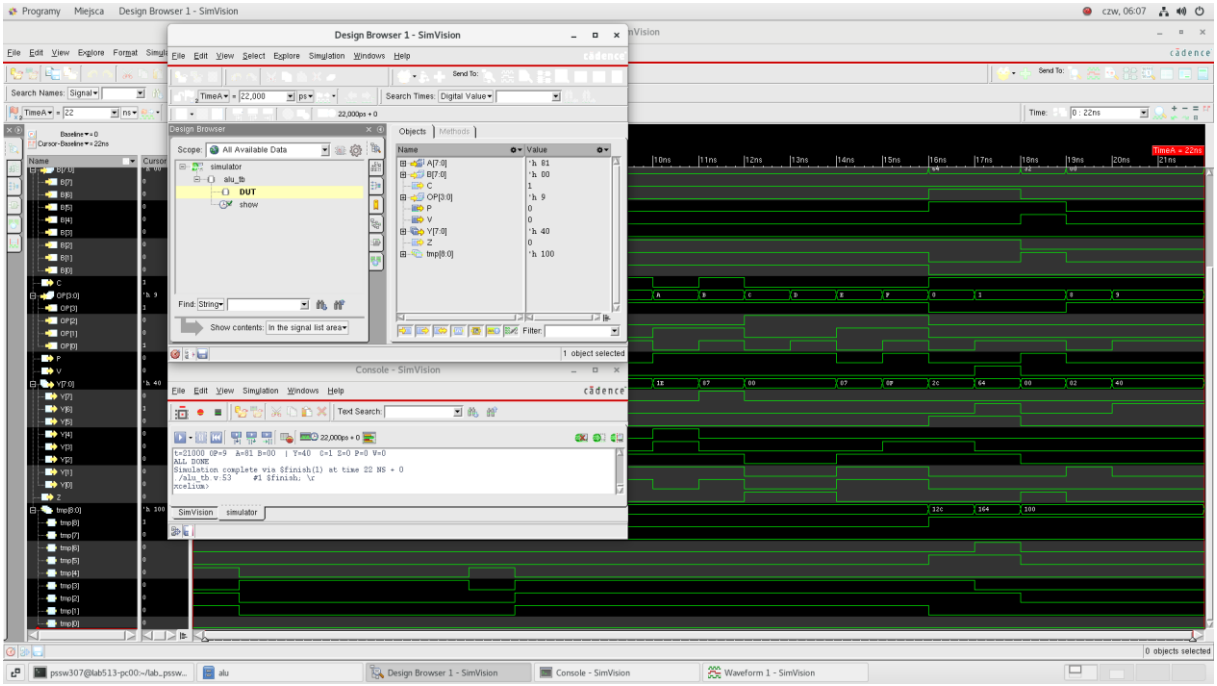
1. ALU

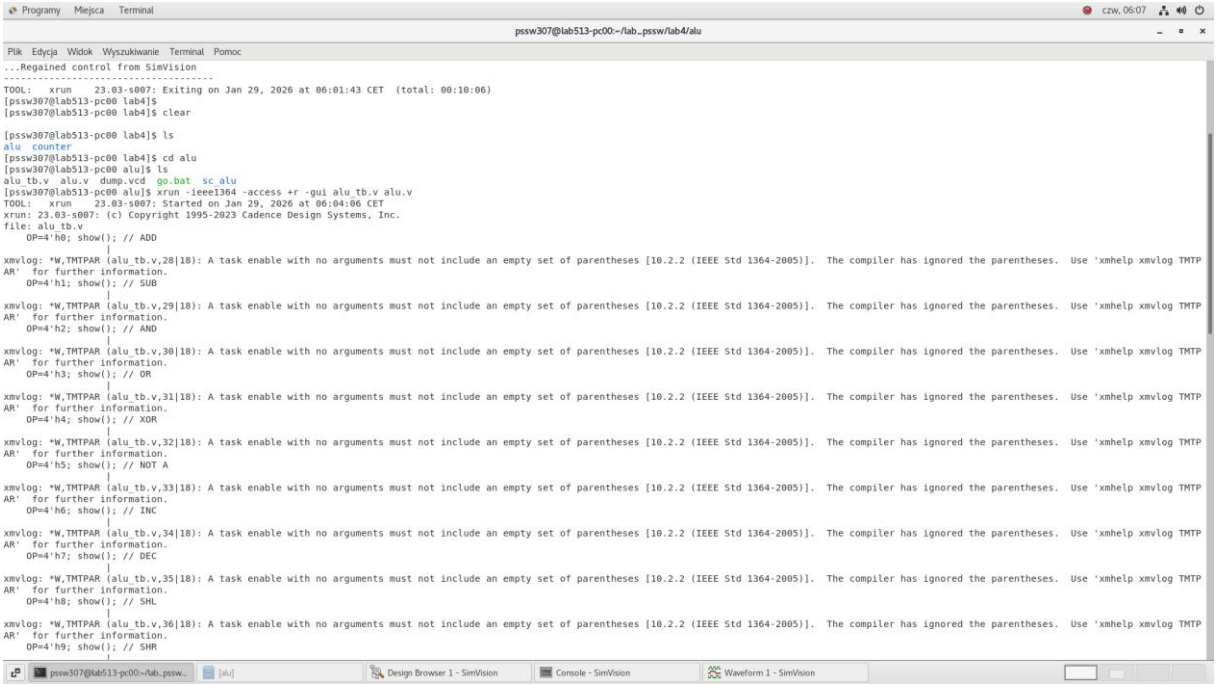
1.1. Symulacja funkcjonalna

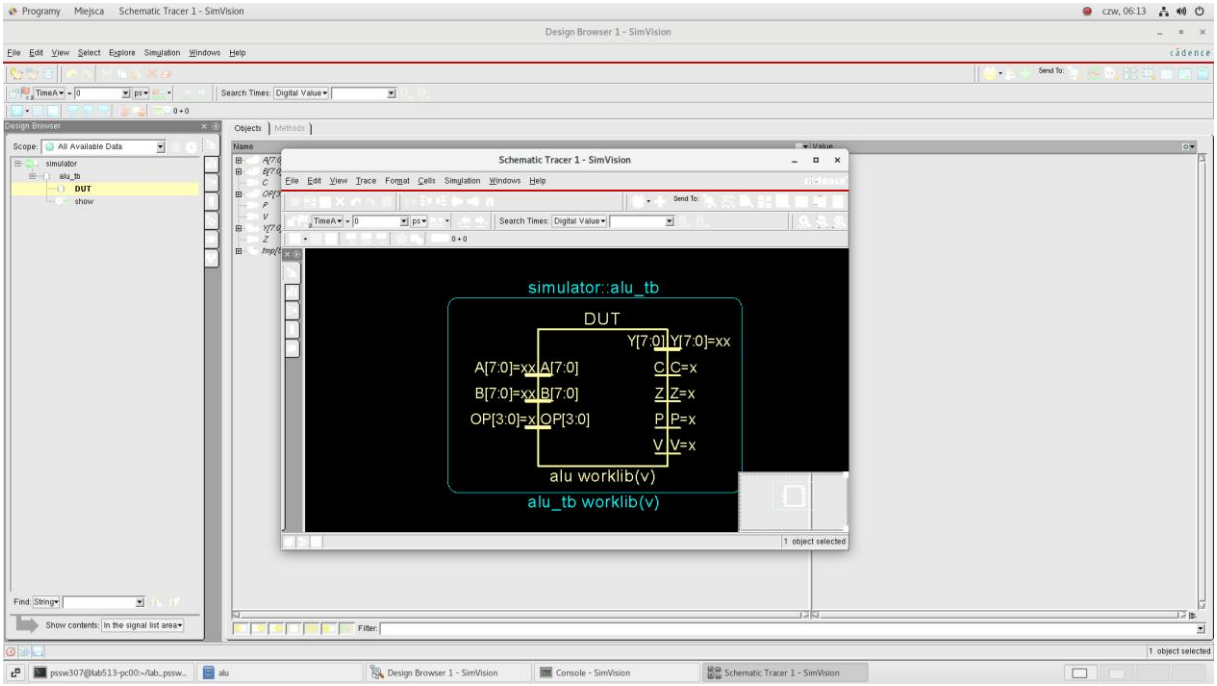



Programy Miejsca Waveform 1 - SimVision
czw, 06:06
Waveform 1 - SimVision
File Edit View Explore Format Simulation Windows Help
cadence
Search Names: Signal
Search Times: Digital Value
Time: 0 : 22ns
Baseline = 0
TimeA = 22ns
Name
Cursor
A[7:0]
B[7:0]
C
OP[3:0]
P
V
Y[7:0]
Z
tmp[8:0]
0 objects selected
psw307@lab513-pc00:~/lab..psw...
alu
Design Browser 1 - SimVision
Console - SimVision
Waveform 1 - SimVision

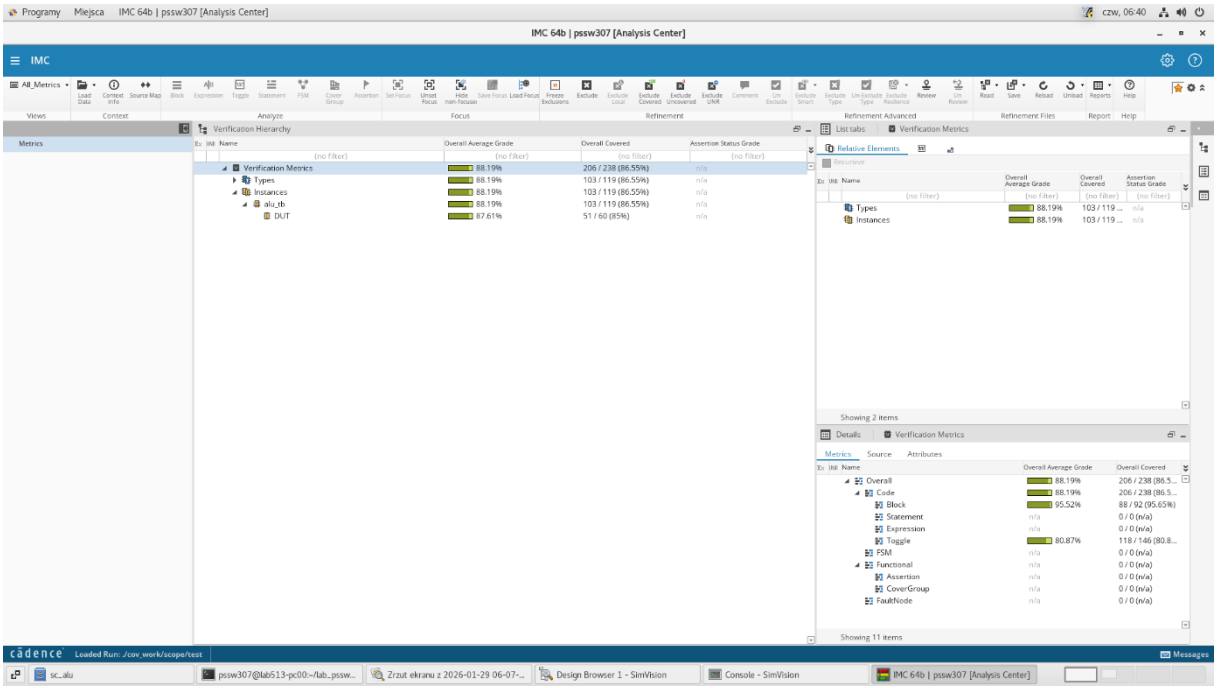

Programy Miejsca Waveform 1 - SimVision
czw, 06:06
Waveform 1 - SimVision
File Edit View Explore Format Simulation Windows Help
cadence
Search Names: Signal
Search Times: Digital Value
Time: 0 : 22ns
Baseline = 0
TimeA = 22ns
Name
Cursor
B[7:0]
C
OP[3:0]
P
V
Y[7:0]
Z
tmp[8:0]
0 objects selected
psw307@lab513-pc00:~/lab..psw...
alu
Design Browser 1 - SimVision
Console - SimVision
Waveform 1 - SimVision

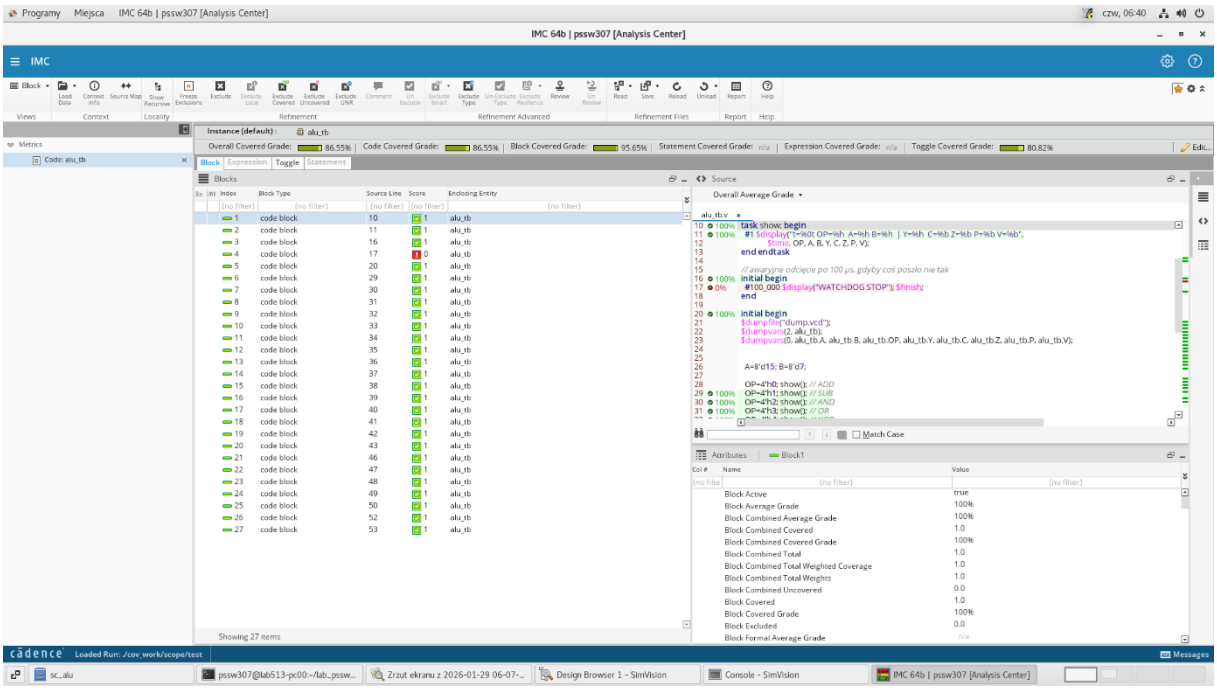

Design Browser 1 - SimVision
Console - SimVision
Waveform 1 - SimVision
DUT
show
simulator
ALL DONE
Simulation complete via $finish(1) at time 22 NS + 0
/alu_tb.v:53 #1 $finish; \c
xcelium>

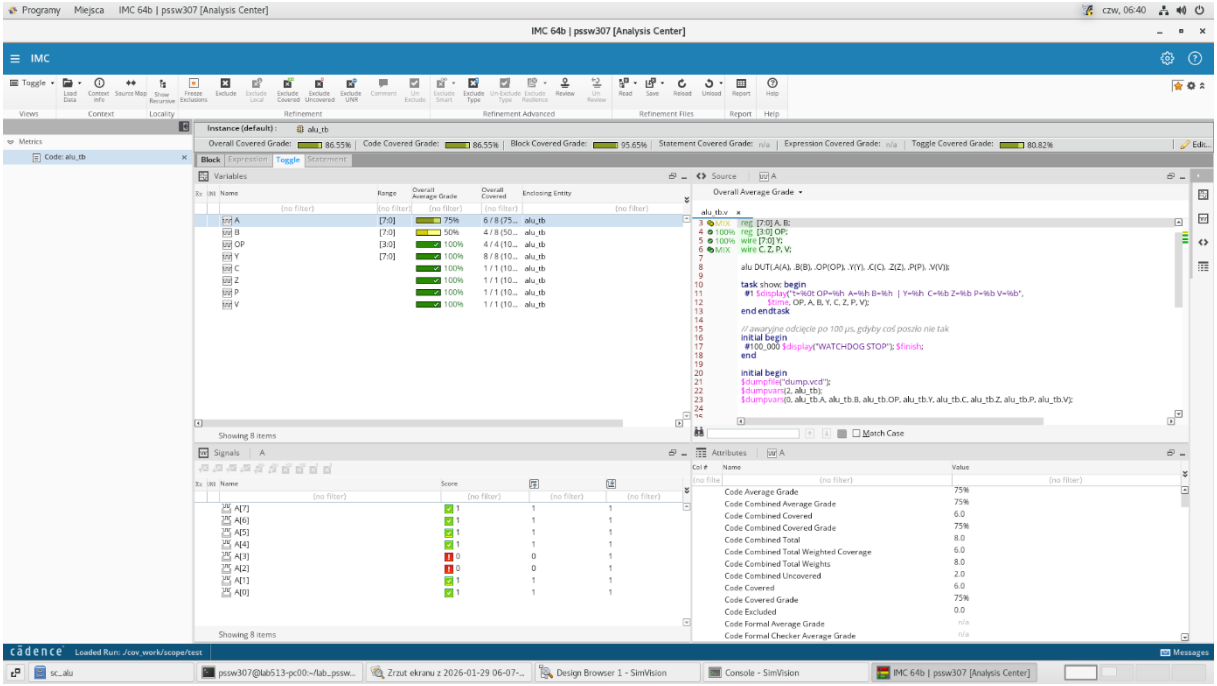

pssw307@lab513-pc00:~/lab_pssw/lab4/alu
...Regained control from SimVision
TOOL: xrun 23.03-s007: Exiting on Jan 29, 2026 at 06:01:43 CET (total: 00:10:06)
[pssw307@lab513-pc00 lab4]$
[pssw307@lab513-pc00 lab4]$ clear
[pssw307@lab513-pc00 lab4]$ ls
alu counter
[pssw307@lab513-pc00 lab4]$ cd alu
[pssw307@lab513-pc00 alu]$ ls
alu_tb.v alu.v dump.vcd go.bat sc_alu
[pssw307@lab513-pc00 alu]$ xrun -ieee1364 -access +r -gui alu_tb.v alu.v
TOOL: xrun 23.03-s007: Started on Jan 29, 2026 at 06:04:06 CET
xrun: 23.03-s007: (c) Copyright 1995-2023 Cadence Design Systems, Inc.
file: alu_tb.v
OP=4'h0; show(); // ADD
xmvlog: *W,TMTPAR (alu_tb.v,28|18): A task enable with no arguments must not include an empty set of parentheses [10.2.2 (IEEE Std 1364-2005)]. The compiler has ignored the parentheses. Use 'xmhelp xmvlog TMTPAR' for further information.
OP=4'h1; show(); // SUB
xmvlog: *W,TMTPAR (alu_tb.v,29|18): A task enable with no arguments must not include an empty set of parentheses [10.2.2 (IEEE Std 1364-2005)]. The compiler has ignored the parentheses. Use 'xmhelp xmvlog TMTPAR' for further information.
OP=4'h2; show(); // AND
xmvlog: *W,TMTPAR (alu_tb.v,30|18): A task enable with no arguments must not include an empty set of parentheses [10.2.2 (IEEE Std 1364-2005)]. The compiler has ignored the parentheses. Use 'xmhelp xmvlog TMTPAR' for further information.
OP=4'h3; show(); // OR
xmvlog: *W,TMTPAR (alu_tb.v,31|18): A task enable with no arguments must not include an empty set of parentheses [10.2.2 (IEEE Std 1364-2005)]. The compiler has ignored the parentheses. Use 'xmhelp xmvlog TMTPAR' for further information.
OP=4'h4; show(); // XOR
xmvlog: *W,TMTPAR (alu_tb.v,32|18): A task enable with no arguments must not include an empty set of parentheses [10.2.2 (IEEE Std 1364-2005)]. The compiler has ignored the parentheses. Use 'xmhelp xmvlog TMTPAR' for further information.
OP=4'h5; show(); // NOT A
xmvlog: *W,TMTPAR (alu_tb.v,33|18): A task enable with no arguments must not include an empty set of parentheses [10.2.2 (IEEE Std 1364-2005)]. The compiler has ignored the parentheses. Use 'xmhelp xmvlog TMTPAR' for further information.
OP=4'h6; show(); // INC
xmvlog: *W,TMTPAR (alu_tb.v,34|18): A task enable with no arguments must not include an empty set of parentheses [10.2.2 (IEEE Std 1364-2005)]. The compiler has ignored the parentheses. Use 'xmhelp xmvlog TMTPAR' for further information.
OP=4'h7; show(); // DEC
xmvlog: *W,TMTPAR (alu_tb.v,35|18): A task enable with no arguments must not include an empty set of parentheses [10.2.2 (IEEE Std 1364-2005)]. The compiler has ignored the parentheses. Use 'xmhelp xmvlog TMTPAR' for further information.
OP=4'h8; show(); // SHL
xmvlog: *W,TMTPAR (alu_tb.v,36|18): A task enable with no arguments must not include an empty set of parentheses [10.2.2 (IEEE Std 1364-2005)]. The compiler has ignored the parentheses. Use 'xmhelp xmvlog TMTPAR' for further information.
OP=4'h9; show(); // SHR



1.2. Pokrycie testów

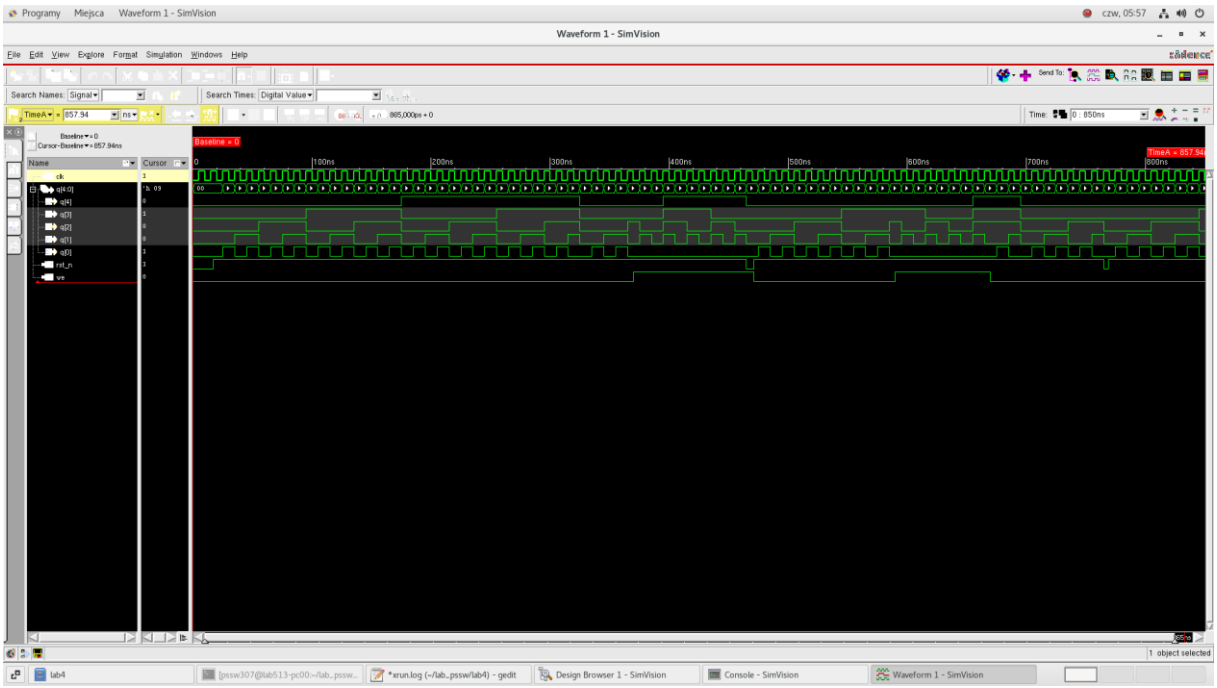


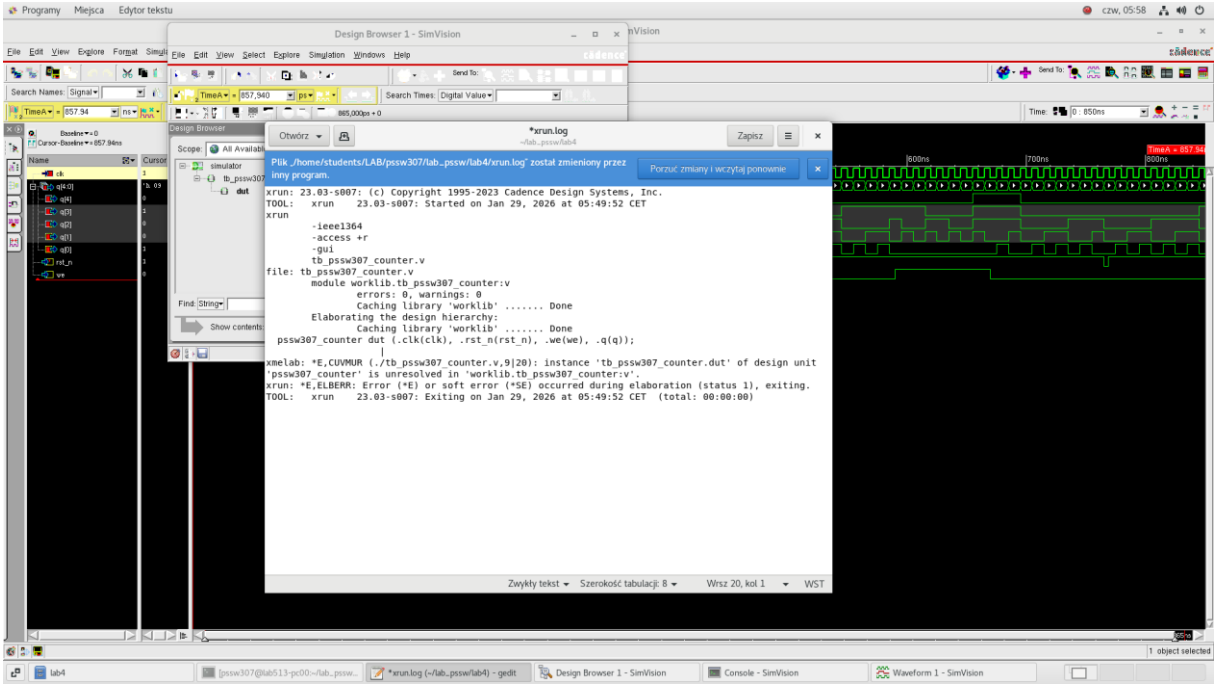


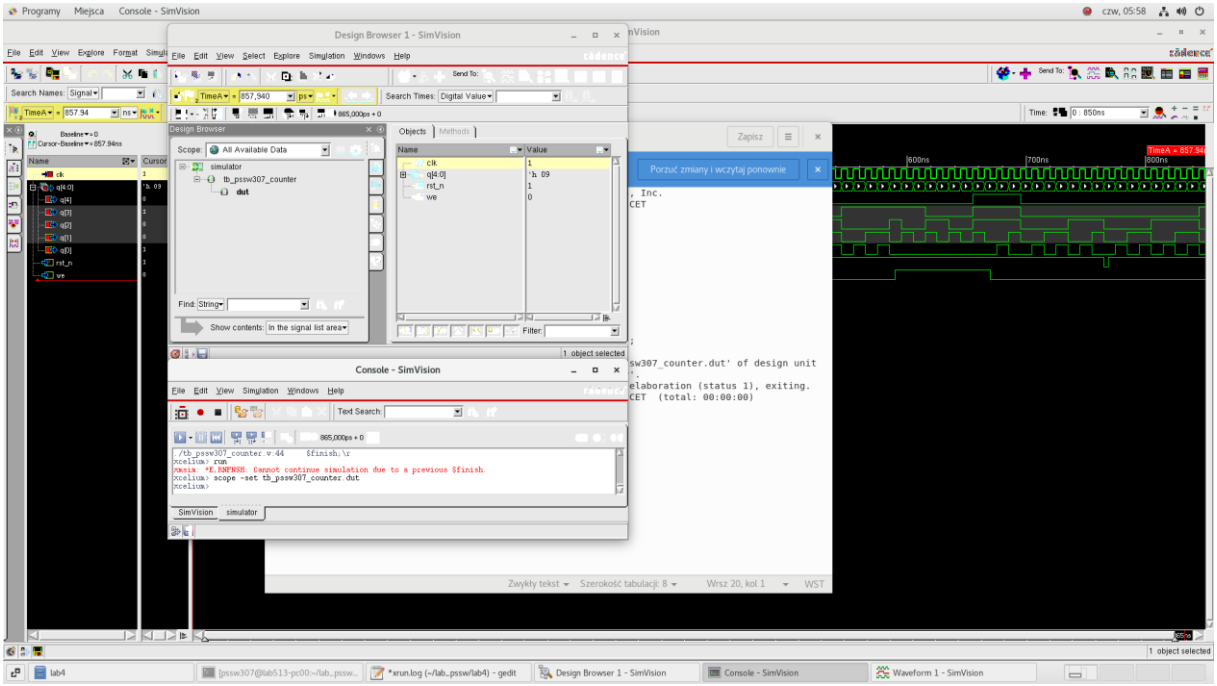


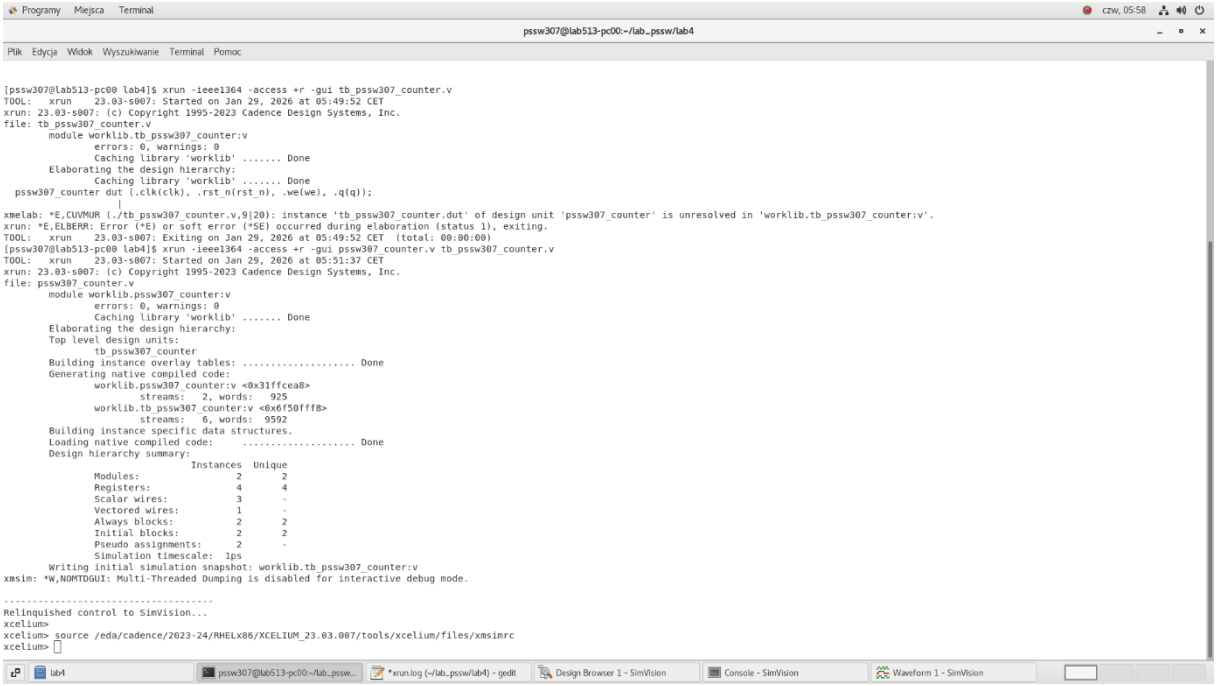
2. Licznik

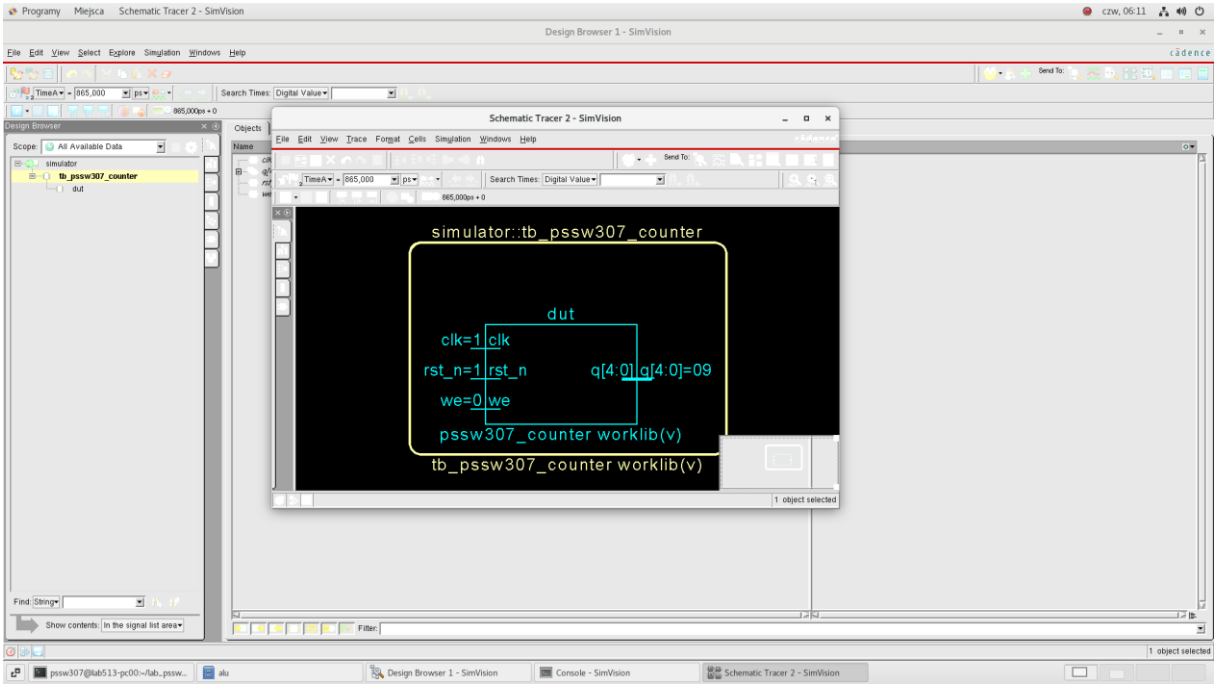
2.1. Symulacja funkcjonalna


Programy Miejsca Waveform 1 - SimVision
czw, 05:57
Waveform 1 - SimVision
File Edit View Explore Format Simulation Windows Help
Search Names: Signal
Search Times: Digital Value
Baseline = 0
TimeA = 857.94
0
100ns
200ns
300ns
400ns
500ns
600ns
700ns
800ns
Name
Cursor
ck
q[4:0]
q[4]
q[3]
q[2]
q[1]
q[0]
rst_n
we
1 object selected
lab4
[pssw307@lab513-pc00:~/lab..pssw...
*xrun.log (~/lab..pssw/lab4) - gedit
Design Browser 1 - SimVision
Console - SimVision
Waveform 1 - SimVision

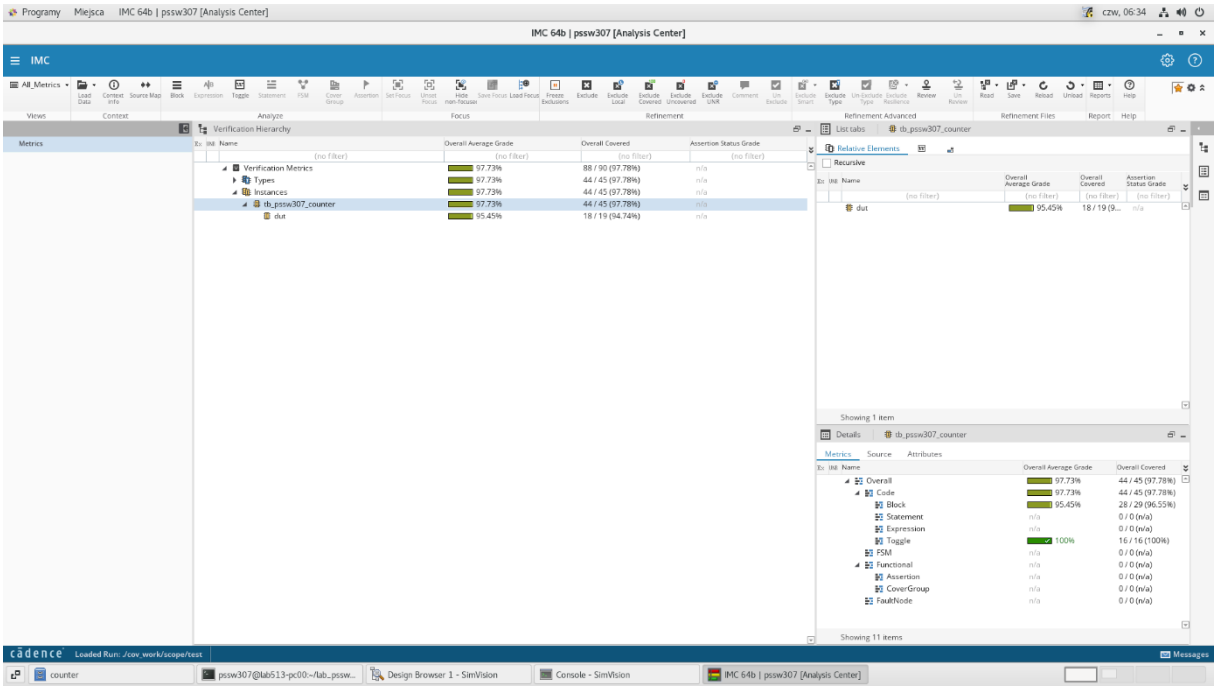

Programy Miejsca Edytor tekstu
czw, 05:58
Design Browser 1 - SimVision
File Edit View Select Explore Simulation Windows Help
Scope: All Available
simulator
tb_pssw307
dut
Find: String
Show contents
*xrun.log
~/lab..pssw/lab4
Otwórz
Zapisz
Plik ./home/students/LAB/pssw307/lab..pssw/lab4/xrun.log' został zmieniony przez inny program.
Porzuć zmiany i wczytaj ponownie
xrun: 23.03-s007: (c) Copyright 1995-2023 Cadence Design Systems, Inc.
TOOL: xrun 23.03-s007: Started on Jan 29, 2026 at 05:49:52 CET
xrun
-ieee1364
-access +r
-gui
tb_pssw307_counter.v
file: tb_pssw307_counter.v
module worklib.tb_pssw307_counter:v
errors: 0, warnings: 0
Caching library 'worklib' ..... Done
Elaborating the design hierarchy:
Caching library 'worklib' ..... Done
pssw307_counter dut (.clk(clk), .rst_n(rst_n), .we(we), .q(q));
xmelab: *E,CUVMUR (./tb_pssw307_counter.v,9|20): instance 'tb_pssw307_counter.dut' of design unit 'pssw307_counter' is unresolved in 'worklib.tb_pssw307_counter:v'.
xrun: *E,ELBERR: Error (*E) or soft error (*SE) occurred during elaboration (status 1), exiting.
TOOL: xrun 23.03-s007: Exiting on Jan 29, 2026 at 05:49:52 CET (total: 00:00:00)
Zwykły tekst
Szerokość tabulacji: 8
Wrsz 20, kol 1
WST
1 object selected
lab4
[pssw307@lab513-pc00:~/lab..pssw...
*xrun.log (~/lab..pssw/lab4) - gedit
Design Browser 1 - SimVision
Console - SimVision
Waveform 1 - SimVision

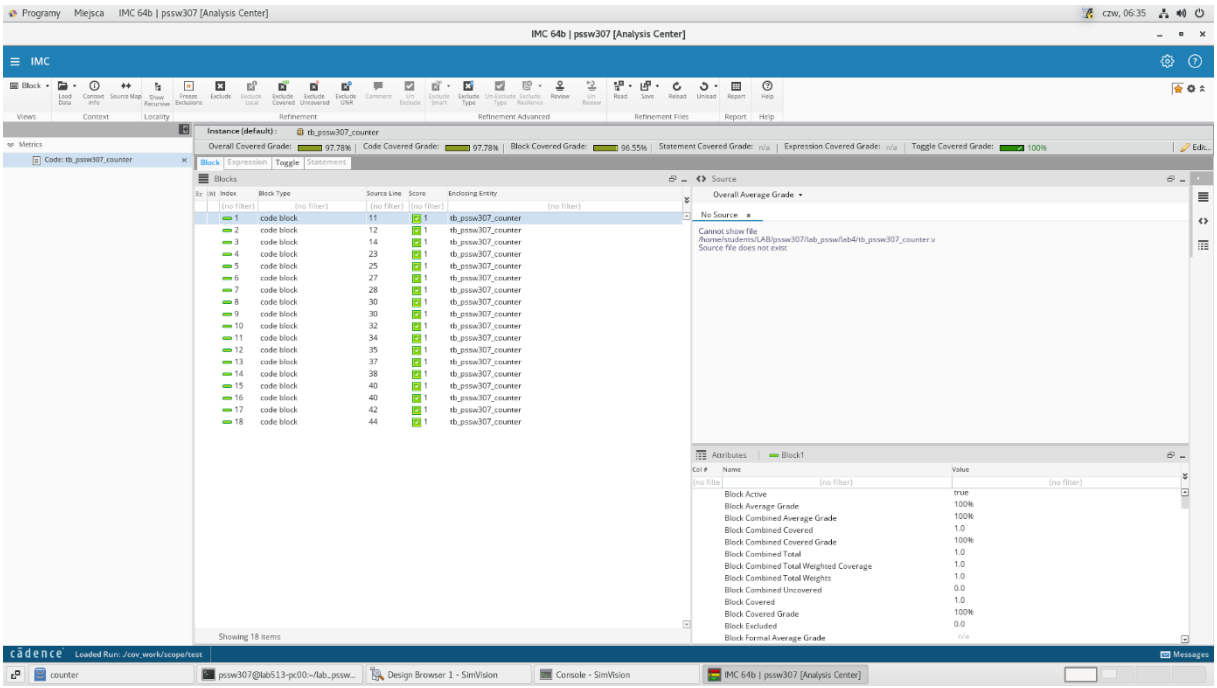

Design Browser 1 - SimVision
Console - SimVision
./tb_pssw307_counter.v:44 $finish,\r
xcelium> run
ncsim: *E,INFINSH: Cannot continue simulation due to a previous $finish.
xcelium> scope -set tb_pssw307_counter.dut
xcelium>
Waveform 1 - SimVision

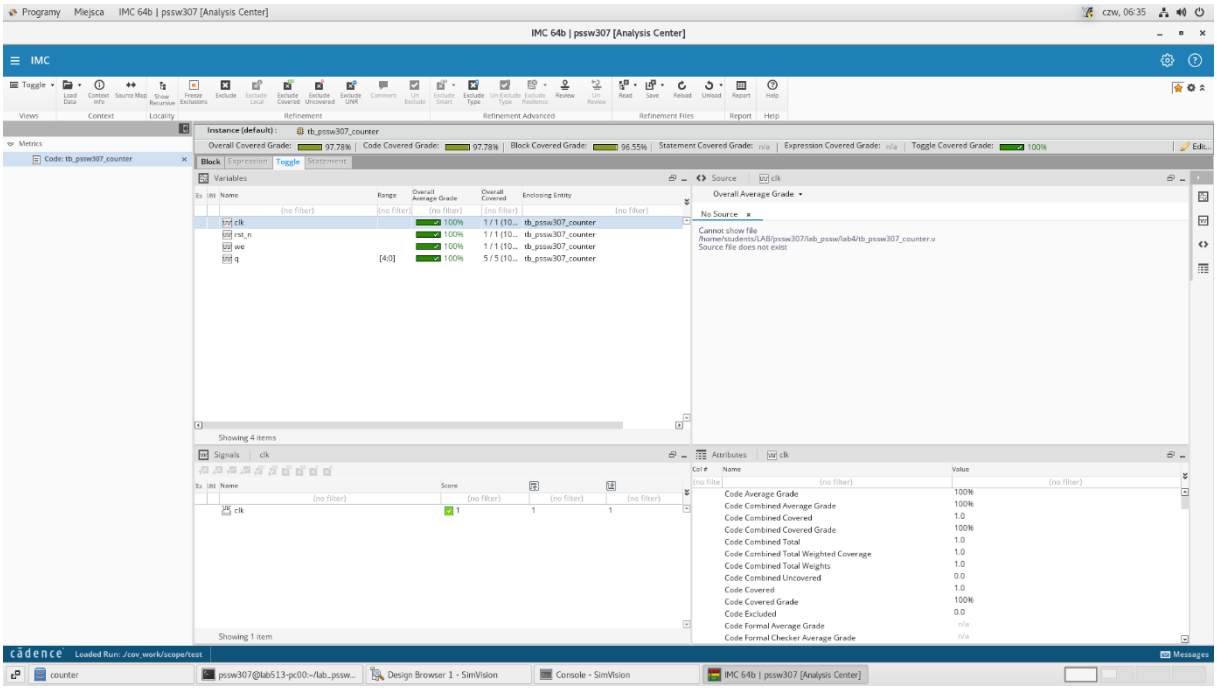

pssw307@lab513-pc00:~/lab_pssw/lab4
[pssw307@lab513-pc00 lab4]$ xrun -ieee1364 -access +r -gui tb_pssw307_counter.v
TOOL: xrun 23.03-s007: Started on Jan 29, 2026 at 05:49:52 CET
xrun: 23.03-s007: (c) Copyright 1995-2023 Cadence Design Systems, Inc.
file: tb_pssw307_counter.v
module worklib.tb_pssw307_counter:v
errors: 0, warnings: 0
Caching library 'worklib' ..... Done
Elaborating the design hierarchy:
Caching library 'worklib' ..... Done
pssw307_counter dut (.clk(clk), .rst_n(rst_n), .we(we), .q(q));
xmelab: *E,CUVMUR (./tb_pssw307_counter.v,9|20): instance 'tb_pssw307_counter.dut' of design unit 'pssw307_counter' is unresolved in 'worklib.tb_pssw307_counter:v'.
xrun: *E,ELBERR: Error (*E) or soft error (*SE) occurred during elaboration (status 1), exiting.
TOOL: xrun 23.03-s007: Exiting on Jan 29, 2026 at 05:49:52 CET (total: 00:00:00)
[pssw307@lab513-pc00 lab4]$ xrun -ieee1364 -access +r -gui pssw307_counter.v tb_pssw307_counter.v
TOOL: xrun 23.03-s007: Started on Jan 29, 2026 at 05:51:37 CET
xrun: 23.03-s007: (c) Copyright 1995-2023 Cadence Design Systems, Inc.
file: pssw307_counter.v
module worklib.pssw307_counter:v
errors: 0, warnings: 0
Caching library 'worklib' ..... Done
Elaborating the design hierarchy:
Top level design units:
tb_pssw307_counter
Building instance overlay tables: ..... Done
Generating native compiled code:
worklib.pssw307_counter:v <0x31ffcea8>
streams: 2, words: 925
worklib.tb_pssw307_counter:v <0x6f50fff8>
streams: 6, words: 9592
Building instance specific data structures.
Loading native compiled code: ..... Done
Design hierarchy summary:
Instances Unique
Modules: 2 2
Registers: 4 4
Scalar wires: 3 -
Vectored wires: 1 -
Always blocks: 2 2
Initial blocks: 2 2
Pseudo assignments: 2 -
Simulation timescale: 1ps
Writing initial simulation snapshot: worklib.tb_pssw307_counter:v
xmsim: *W,NOMTDGUI: Multi-Threaded Dumping is disabled for interactive debug mode.
Relinquished control to SimVision...
xcelium>
xcelium> source /eda/cadence/2023-24/RHELx86/XCELIUM_23.03.007/tools/xcelium/files/xmsimrc
xcelium>



2.2. Pokrycie testów




IMC 64b | pssw307 [Analysis Center]
Instance (default) : tb_pssw307_counter
Overall Covered Grade: 97.78% | Code Covered Grade: 97.78% | Block Covered Grade: 96.55% | Statement Covered Grade: n/a | Expression Covered Grade: n/a | Toggle Covered Grade: 100%
Blocks
Showing 18 items
Source
No Source
Cannot show file /home/students/LAB/pssw307/lab_pssw/lab4/tb_pssw307_counter.v
Source file does not exist
Attributes
Block1


IMC 64b | pssw307 [Analysis Center]
Instance (default) : tb_pssw307_counter
Overall Covered Grade: 97.78% | Code Covered Grade: 97.78% | Block Covered Grade: 96.55% | Statement Covered Grade: n/a | Expression Covered Grade: n/a | Toggle Covered Grade: 100%
Variables
Showing 4 items
Source
No Source
Cannot show file /home/students/LAB/pssw307/lab_pssw/lab4/tb_pssw307_counter.v
Source file does not exist
Signals
Showing 1 item
Attributes